

# Multi-window MapBiomass — territory vs buffer summary

Run: `yvynation_batch_20260525_1655`. Step = 8 years, stages 1985 → 1993 → 2001 → 2009 → 2017 → 2024. Buffer = 10 km external ring. Only the **multi-time-window MapBiomass** analysis was enabled, so the comparison rests entirely on the long-format transitions CSVs.

All deltas are in percentage **points** of each region's own total area (which is held constant by design). "pp" = percentage points.

## Bacurizinho (4901) — Maranhão

| Region       | Total      | Forest 1985 | Forest 2024  | Forest Δ                        | Major 2024 gainers                                                       |
|--------------|------------|-------------|--------------|---------------------------------|--------------------------------------------------------------------------|
| Territory    | 83,511 ha  | 99.6%       | 98.3%        | <b>−1.3 pp</b><br>(−1,054 ha)   | tiny Forest plantation (0.7%), Mosaic (+0.2 pp)                          |
| Buffer 10 km | 150,800 ha | 84.4%       | <b>58.9%</b> | <b>−25.5 pp</b><br>(−38,462 ha) | Pasture +11.8 pp, Mosaic +6.4 pp, Forest plantation +6.8 pp, Soy +3.1 pp |

**Difference:** the protected area is essentially intact (≈99% forest, no agriculture), while its 10-km surroundings lost a quarter of forest cover, replaced mainly by pasture, mosaic-of-uses and new agro-forestry (soy + planted forest).

## Bacurizinho (4902) — expansion polygon

| Region       | Total      | Forest 1985 | Forest 2024  | Forest Δ                        | Major 2024 gainers                                                      |
|--------------|------------|-------------|--------------|---------------------------------|-------------------------------------------------------------------------|
| Territory    | 135,599 ha | 86.2%       | 80.0%        | <b>−6.2 pp</b><br>(−8,390 ha)   | Pasture +5.8 pp, Mosaic +2.5 pp                                         |
| Buffer 10 km | 192,682 ha | 71.2%       | <b>49.2%</b> | <b>−22.0 pp</b><br>(−42,359 ha) | Pasture +9.3 pp, Forest plantation +7.5 pp, Mosaic +6.2 pp, Soy +2.7 pp |

**Difference:** the expansion polygon is itself under noticeable pressure — pasture/mosaic now occupy ~8% of it — but the buffer is converting roughly **3.5× faster** in absolute terms (42 kha vs 8 kha). Savanna also halves outside (−8.5 kha) while it only drops ~3.6 kha inside.

## Betânia — Amazonian floodplain TI

| Region          | Total         | Forest+Floodable<br>1985 | Forest+Floodable<br>2024 | Δ                                        | Notes                                                |
|-----------------|---------------|--------------------------|--------------------------|------------------------------------------|------------------------------------------------------|
| Territory       | 123,343<br>ha | 90.8%                    | 89.2%                    | <b>-1.6 pp</b><br>(Floodable<br>-1.4 pp) | River area<br>essentially stable;<br>Wetland +1.8 pp |
| Buffer 10<br>km | 245,344<br>ha | 93.1%                    | 91.7%                    | <b>-1.4 pp</b><br>(Floodable<br>-1.2 pp) | Tiny pasture<br>(0.3%) and urban<br>(0.1%) appearing |

**Difference:** unique among the set — territory and buffer behave **almost identically**. This is a river-dominated Amazonian landscape where the surroundings are still mostly intact forest, so the buffer is not a frontier here. The +1.8 pp wetland gain inside the TI is the strongest signal, likely flooded forest converting to open wetland.

Kayapó NW

| Region          | Total         | Forest<br>1985 | Forest<br>2024 | Forest Δ                        | Major 2024 gainers              |
|-----------------|---------------|----------------|----------------|---------------------------------|---------------------------------|
| Territory       | 857,434<br>ha | 93.8%          | 93.6%          | <b>-0.2 pp</b> (-1,649 ha)      | none significant                |
| Buffer 10<br>km | 210,425<br>ha | 88.1%          | <b>66.4%</b>   | <b>-21.7 pp</b> (-45,576<br>ha) | Pasture +23.7 pp (50,703<br>ha) |

**Difference:** the territory's NW quadrant is essentially undisturbed; the buffer is being cleared into pasture at a high rate — pasture went from <1% to 24% in ~40 years.

Kayapó NE — the hot spot

| Region          | Total           | Forest<br>1985 | Forest<br>2024 | Forest Δ                         | Major 2024 gainers                               |
|-----------------|-----------------|----------------|----------------|----------------------------------|--------------------------------------------------|
| Territory       | 1,098,697<br>ha | 84.9%          | 83.9%          | <b>-0.9 pp</b> (-10,206<br>ha)   | Mining +0.7 pp (8,272 ha)                        |
| Buffer 10<br>km | 161,027 ha      | 91.0%          | <b>25.8%</b>   | <b>-65.2 pp</b><br>(-105,022 ha) | Pasture +66.1 pp (108,572<br>ha), Mining +2.1 pp |

**Difference:** by far the most dramatic asymmetry in the whole batch. The TI's NE quadrant lost just 0.9 pp of forest but accumulated 8 kha of internal **mining** (the largest mining footprint of any quadrant or buffer in the run). The buffer collapsed from 91% forest to 26%, with two-thirds now pasture and another 2% mining. The buffer ring around NE is essentially a fully converted agricultural frontier.

Kayapó SE

| Region       | Total      | Forest 1985 | Forest 2024 | Forest Δ              | Major 2024 gainers                           |
|--------------|------------|-------------|-------------|-----------------------|----------------------------------------------|
| Territory    | 691,068 ha | 87.1%       | 85.9%       | -1.2 pp (-8,585 ha)   | Mining +1.4 pp (10,181 ha)                   |
| Buffer 10 km | 164,618 ha | 87.9%       | 56.6%       | -31.3 pp (-51,530 ha) | Pasture +31.1 pp (60,131 ha), Mining +0.8 pp |

**Difference:** another classic pattern — internal forest mostly preserved, but with the second-largest mining footprint inside any quadrant (10.2 kha, slightly more than NE). The buffer lost ~one-third of its forest to pasture.

Kayapó SW — almost frozen

| Region       | Total      | Forest 1985 | Forest 2024 | Forest Δ         | Major 2024 gainers |
|--------------|------------|-------------|-------------|------------------|--------------------|
| Territory    | 685,737 ha | 88.0%       | 88.0%       | ±0.0 pp (+96 ha) | none               |
| Buffer 10 km | 182,899 ha | 70.1%       | 70.1%       | ±0.0 pp (+29 ha) | none               |

**Difference:** essentially no change in either region. SW is far from the deforestation frontier — both the territory and the buffer have stayed stable over 40 years. River area drops 0.3 pp (-587 ha) in the buffer, the only meaningful signal.

Krenak de Sete Salões — *Atlantic Forest, Minas Gerais*

| Region       | Total      | Forest 1985 | Forest 2024 | Forest Δ            | Major dynamic                                            |
|--------------|------------|-------------|-------------|---------------------|----------------------------------------------------------|
| Territory    | 16,607 ha  | 43.4%       | 49.8%       | +6.4 pp (+1,065 ha) | Pasture -4.2 pp, Mosaic -4.9 pp — <b>forest regrowth</b> |
| Buffer 10 km | 109,104 ha | 14.4%       | 14.6%       | +0.2 pp             | Pasture -1.2 pp, Coffee +1.0 pp (+1,121 ha)              |

**Difference:** the only **forest-gaining** territory in the batch — a clear inversion of the Amazonian pattern. The TI is recovering Atlantic Forest at the expense of pasture and mosaic, while its buffer remains heavily pastoral (62% pasture) and is mostly substituting older pasture/mosaic with **coffee plantations**.

Krenak — *also Atlantic Forest, MG*

| Region       | Total     | Forest<br>1985 | Forest<br>2024 | Forest $\Delta$          | Major dynamic                                  |
|--------------|-----------|----------------|----------------|--------------------------|------------------------------------------------|
| Territory    | 4,279 ha  | 10.3%          | <b>15.5%</b>   | <b>+5.1 pp</b> (+220 ha) | Pasture -9.0 pp (-385 ha),<br>Mosaic +3.2 pp   |
| Buffer 10 km | 61,276 ha | 17.5%          | 17.2%          | <b>-0.3 pp</b>           | Pasture +2.3 pp (+1,416 ha),<br>Mosaic -3.0 pp |

**Difference:** same direction as Sete Salões but smaller area. The TI is shedding pasture and slowly regaining forest (+5.1 pp); the buffer instead is **intensifying pasture** (+1.4 kha) at the expense of mosaic. Inside vs outside the protected line, the *direction* of change is opposite.

## Cross-cutting takeaways

- **Amazonian TIs (Bacurizinho, Kayapó NW/NE/SE) act as forest enclaves.** Inside they lose <2 pp of forest in 40 years; outside the 10-km ring they lose 20–65 pp, almost entirely to pasture (with soy and planted-forest appearing in Bacurizinho).
- **Kayapó NE/SE buffer rings are completely cleared frontiers** (Forest  $\leq 57\%$  in 2024, pasture  $\geq 30\%$ ); NW is closer to a half-cleared frontier; SW is essentially undisturbed.
- **Mining is an internal — not buffer — threat for Kayapó** (NE: +8.3 kha, SE: +10.2 kha inside the TI). The buffers see only modest mining growth.
- **Betânia behaves as an island in still-forested floodplain:** territory and buffer evolve in parallel, both losing ~1.5 pp of Floodable Forest. There is no clear protective effect because there is no pressure yet.
- **Krenak and Krenak de Sete Salões reverse the pattern.** Both Atlantic-Forest TIs are gaining forest (+5.1 to +6.4 pp) while their buffers stagnate or intensify pasture/coffee — consistent with Cerrado/Mata-Atlântica regrowth dynamics where the deforestation front passed decades earlier.